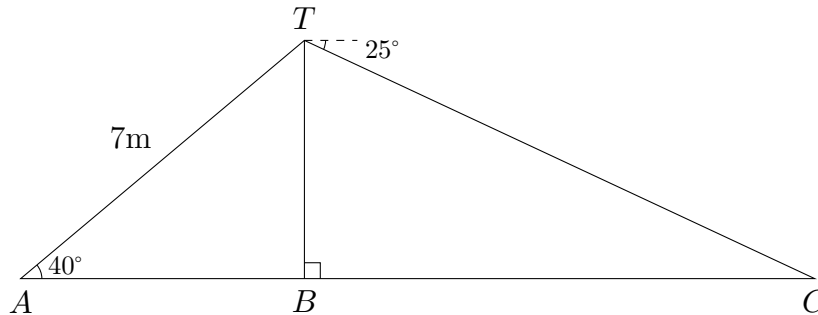


Standard 2 Trigonometry

Question Bank

Question X

A pole is being held up by two guy wires anchored at A and C . The wire connecting A to T is 7m long, and the angle of elevation of T from A is 40° . The angle of depression of C from T is 25° .

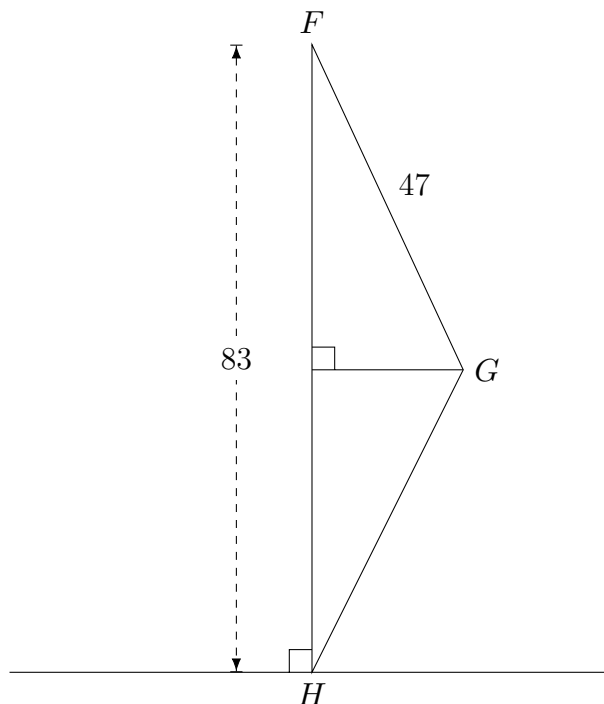


How far apart are A and C ?

Question 1

The triangle FGH is shown below. The length of FG is 47 and the length of FH is 83.

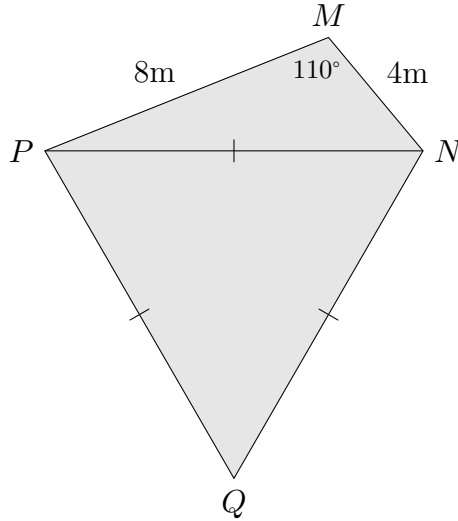
The angle of elevation of G from H is 64° and $\angle FGH$ is obtuse.



Find the length of GH correct to two decimal places.

Question 2

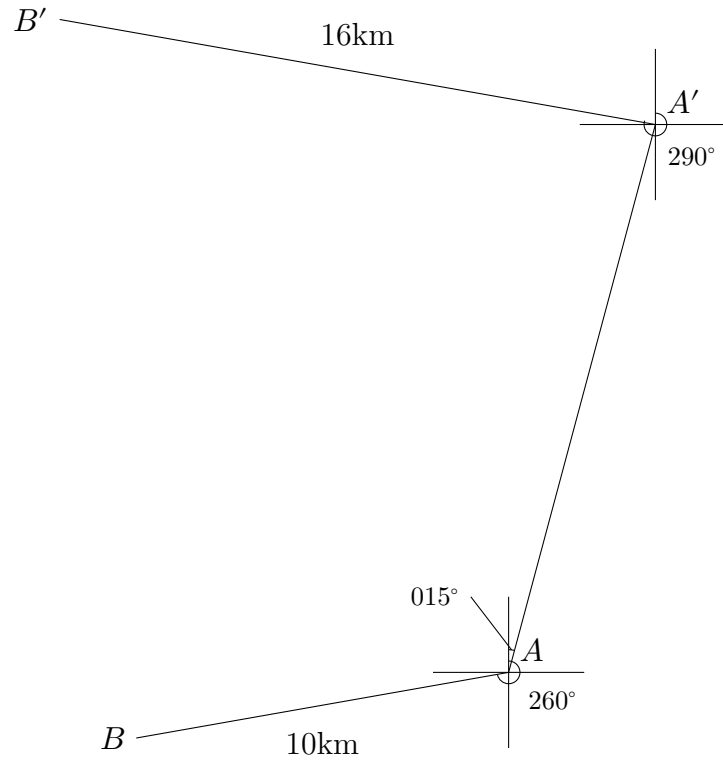
A strange window is constructed from two triangles, PMN and PNQ . The length of PM is 8m and the length of MN is 4m. The $\angle PMN$ is 110° .



The window is to be coated in a protective film that costs $\$45/\text{m}^2$, how much will it cost to coat the window?

Question 3

When Ship 1 is location at position A it determines that Ship 2 is 10km away on a bearing of 260° at point B . Then Ship 1 sails at a speed of 5km/hr on a bearing of 15° for 3 hours, reaching the point A' . Then it made another measurement and found that Ship 2 was 16km away on a bearing of 290° at point B' .



How fast is Ship 2 moving? (Extension question is to find the bearing Ship 2 is travelling on).

Answers

Question 1

Since $\angle FHG$ is complementary to 64° ,

$$\angle FGH = 26^\circ.$$

Find $\angle FGH$ using sine rule,

$$\begin{aligned}\frac{\sin(\angle FGH)}{83} &= \frac{\sin(26^\circ)}{47} \\ \sin(\angle FGH) &= \frac{\sin(26^\circ)}{47} \times 83 \\ \angle FGH &\approx 50.7^\circ\end{aligned}$$

But since $\angle FGH$ is obtuse,

$$\angle FGH \approx 180^\circ - 50.7^\circ = 129.3^\circ$$

$$\angle GFH = 180^\circ - 129.3^\circ - 26^\circ = 24.7^\circ$$

$$\begin{aligned}\frac{GH}{\sin(24.7^\circ)} &= \frac{47}{\sin(26^\circ)} \\ GH &= \frac{47}{\sin(26^\circ)} \times \sin(24.7^\circ) \\ &\approx 44.80\end{aligned}$$

Question 2

The area of $\triangle PMN$ is

$$A_{\triangle PMN} = \frac{1}{2} \times 4 \times 8 \times \sin(110^\circ) \approx 15.04\text{m}^2$$

The length of PN

$$\begin{aligned}PN^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos(110^\circ) \\ PN &= \sqrt{101.889...} \\ &\approx 10.1\text{m}\end{aligned}$$

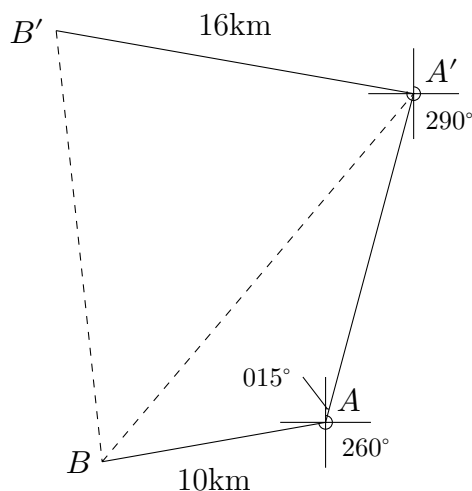
The area of equilateral $\triangle PNQ$

$$A_{\triangle PNQ} = \frac{1}{2} \times 10.1^2 \times \sin(60^\circ) \approx 44.17\text{m}^2$$

The cost is

$$(15.04 + 44.17) \times 45 = \$2664.45$$

Question 3



The length of AA' is $5 \times 3 = 15$.

Find the angle $\angle BAA'$, $360^\circ - 260^\circ + 15^\circ = 115^\circ$.

Use the cosine rule to find the length of $A'B$,

$$\begin{aligned} A'B &= \sqrt{10^2 + 15^2 - 2 \times 10 \times 15 \times \cos(115^\circ)} \\ &\approx 21.26 \end{aligned}$$

Find $\angle BA'A$ using cosine rule

$$\begin{aligned} \angle BA'A &= \cos^{-1} \left(\frac{15^2 + 21.26^2 - 10^2}{2 \times 15 \times 21.26} \right) \\ &\approx 25.22^\circ \end{aligned}$$

Find the angle of $B'A'B$ is $290^\circ - 180^\circ - 15^\circ - 25.22^\circ = 69.78^\circ$

Use the cosine rule to find BB' ,

$$\begin{aligned} B'B &= \sqrt{16^2 + 21.26^2 - 2 \times 16 \times 21.26 \times \cos(69.78^\circ)} \\ &\approx 21.75 \end{aligned}$$

The speed of Ship 2 is $21.75 \div 3 = 7.25\text{km/hr}$.